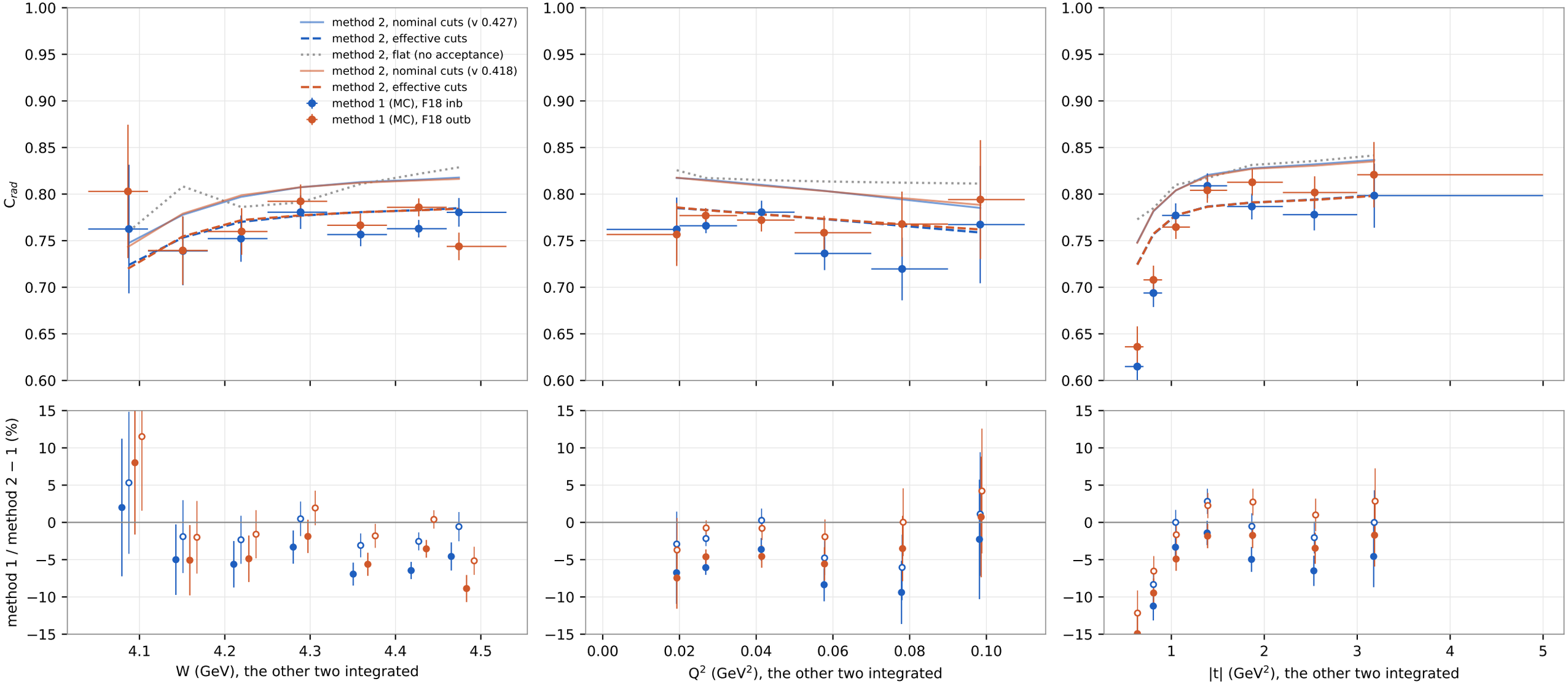
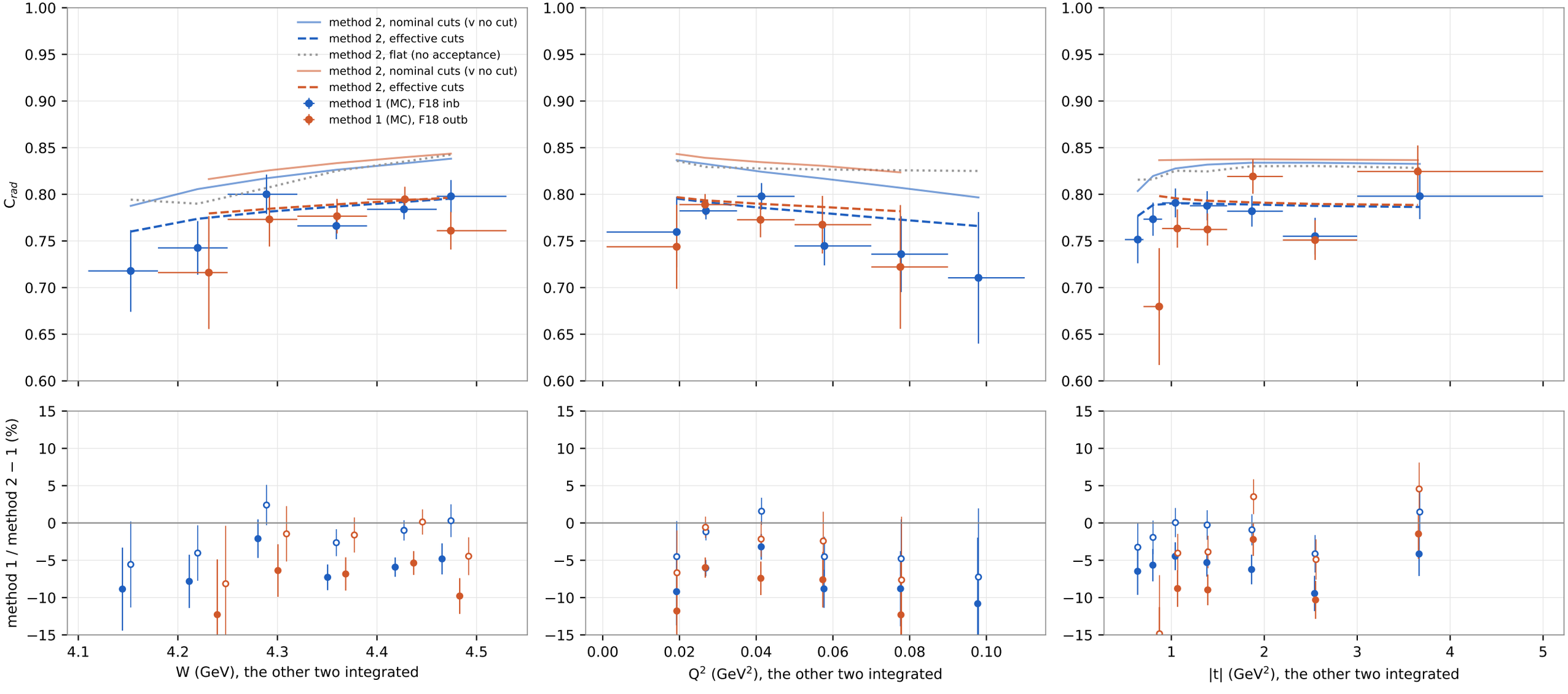
e'e$^+$e$^-$ (proton missing). M. Tenorio's selection: FT e' (θ_e 2.45-6.45°), ν>8.1, lepton PID, |p|>1.95, vz on the inward lepton, 2.95<M(ee)<3.20; 3 tracks: Δp_z>-2.5, MX within 4σ; 4 tracks: proton, -0.7<Δp_z<0.5, |MX^2|<0.036

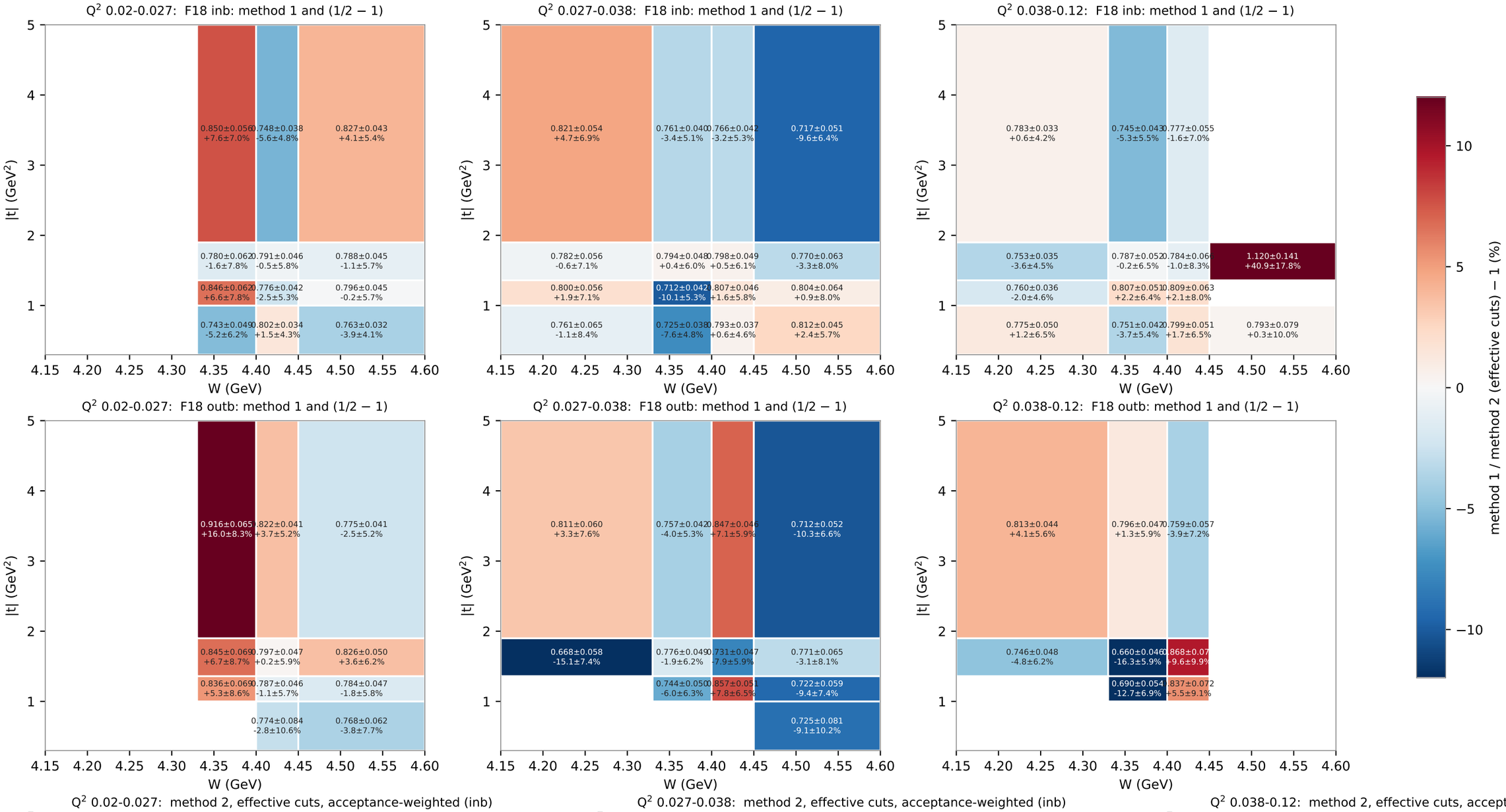


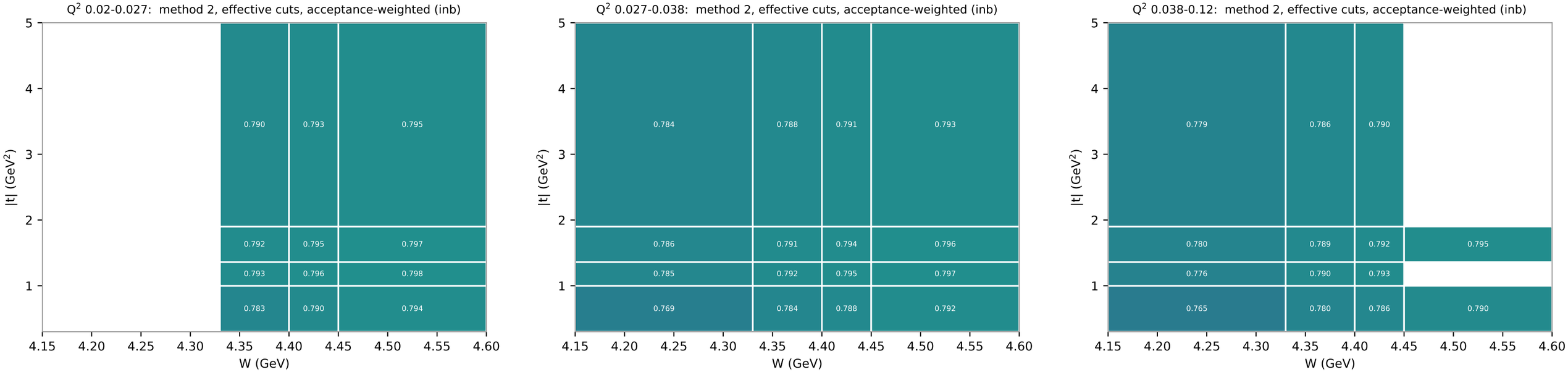
e'e$^+$e$^-$p' (all detected). M. Tenorio's selection: FT e' (θ_e 2.45-6.45°), ν>8.1, lepton PID, |p|>1.95, vz on the inward lepton, 2.95<M(ee)<3.20; 3 tracks: Δp_z>-2.5, MX within 4σ; 4 tracks: proton, -0.7<Δp_z<0.5, |MX^2|<0.036

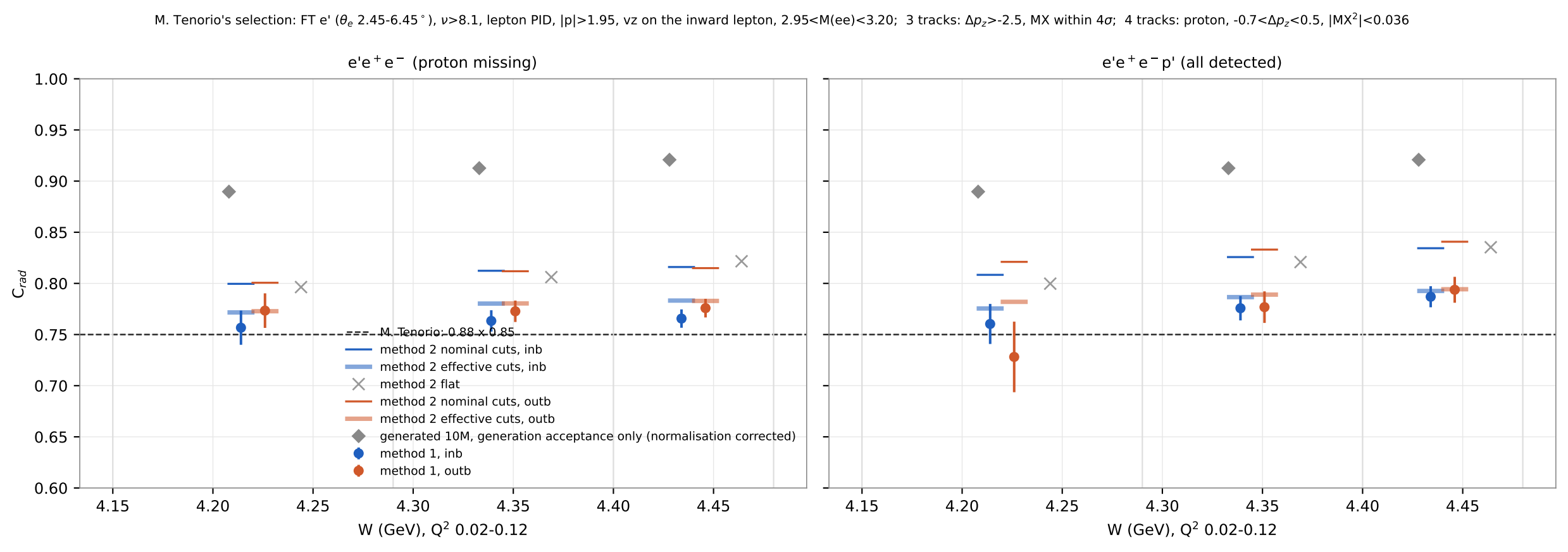


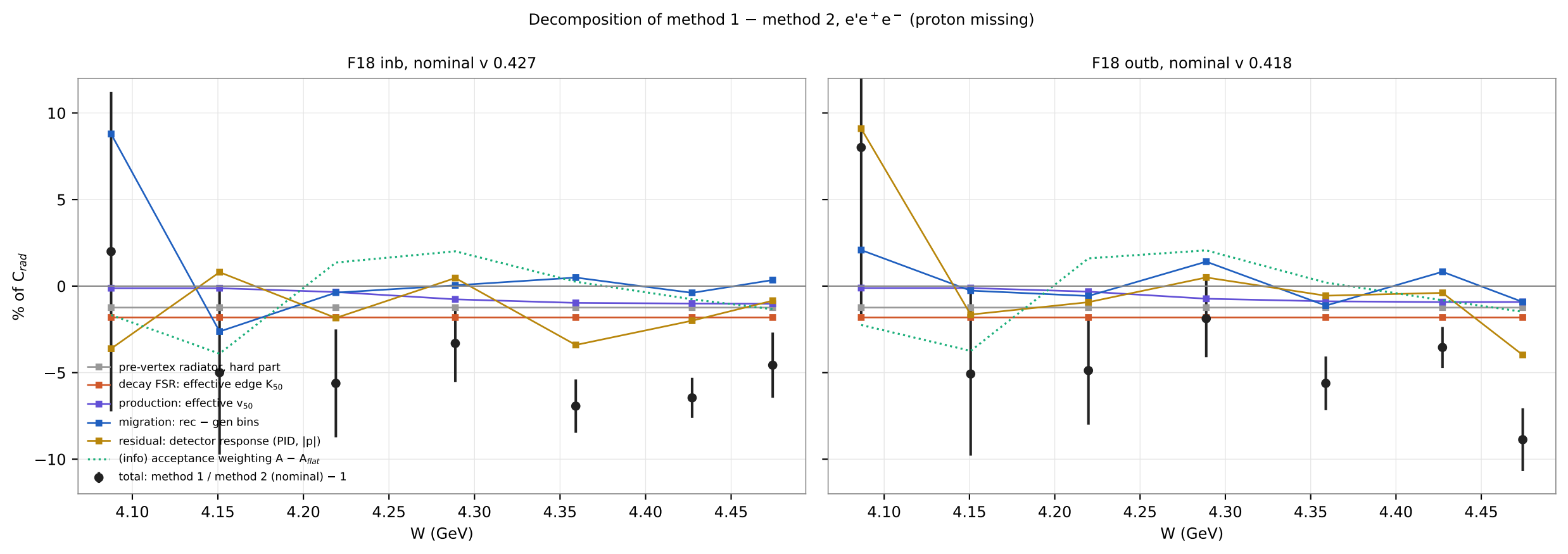
C_{rad}(W, |t|), e'e$^+$e$^-$ (proton missing). Upper rows: method-1 value ± stat, and its deviation from method 2. M. Tenorio's selection: FT e' (θ_e 2.45-6.45°), ν>8.1, lepton PID, |p|>1.95, vz on the inward lepton, 2.95<M(ee)<3.20; 3 tracks: Δp_z>-2.5, MX within 4σ; 4 tracks: proton, -0.7<Δp_z<0.5, |MX^2|<0.036

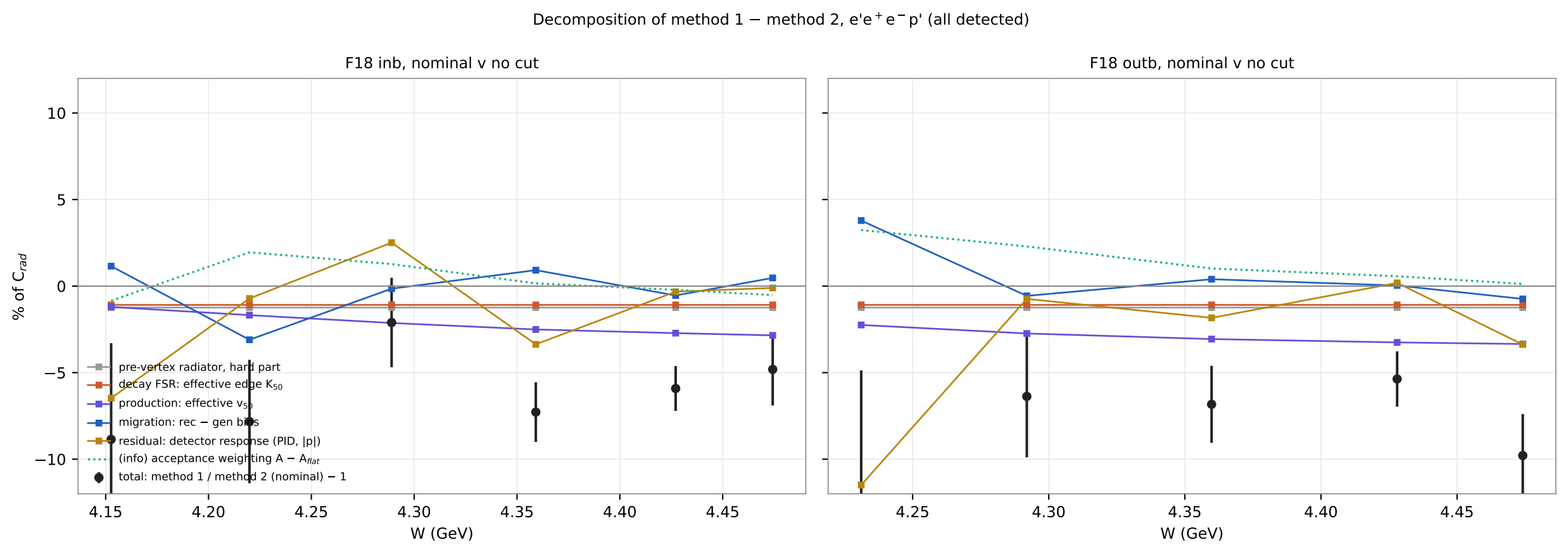
C_{rad}(W, |t|), e'e$^+$e$^-$p' (all detected). Upper rows: method-1 value ± stat, and its deviation from method 2. M. Tenorio's selection: FT e' (θ_e 2.45-6.45°), ν>8.1, lepton PID, |p|>1.95, vz on the inward lepton, 2.95<M(ee)<3.20; 3 tracks: Δp_z>-2.5, MX within 4σ; 4 tracks: proton, -0.7<Δp_z<0.5, |MX^2|<0.036



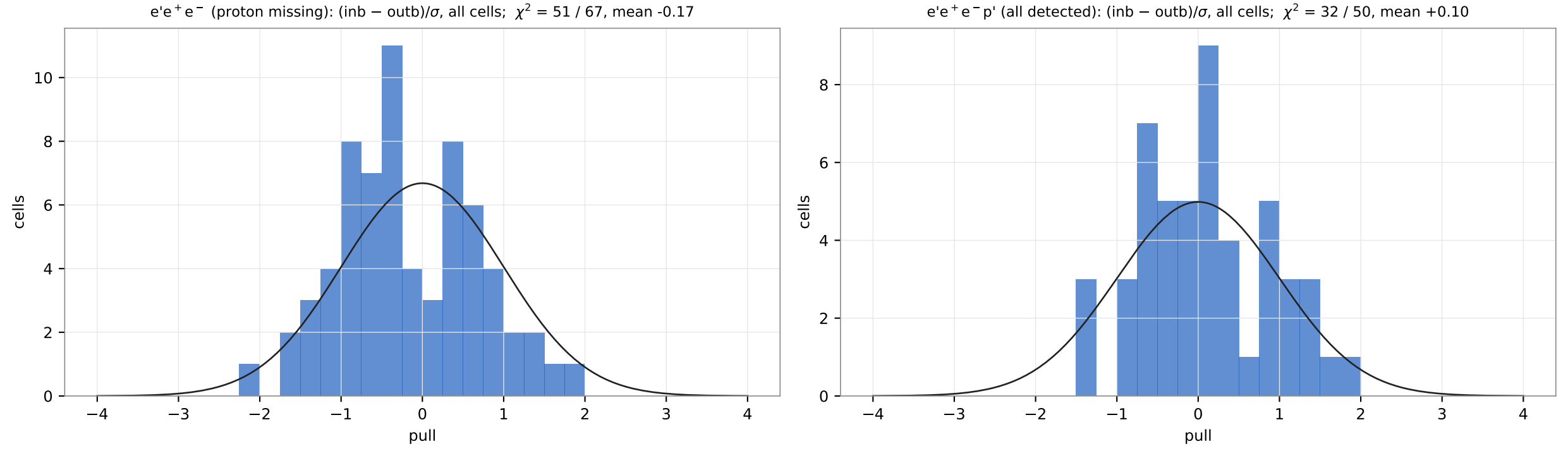


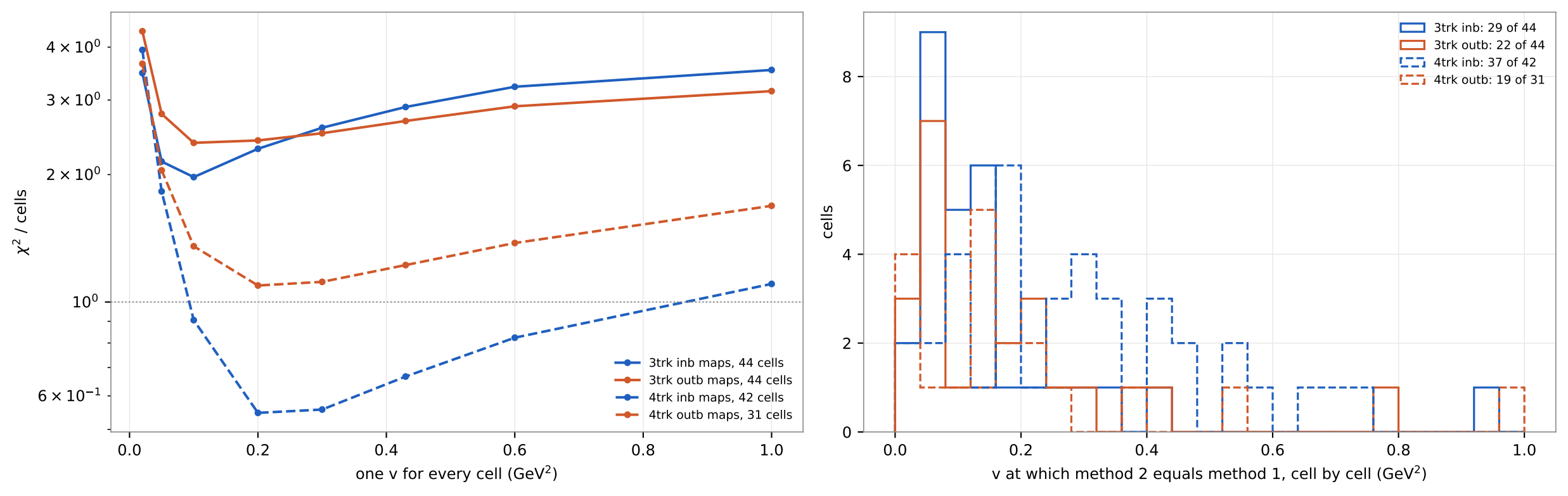

M. Tenorio's selection: FT e' (θ_e 2.45-6.45°), ν>8.1, lepton PID, |p|>1.95, vz on the inward lepton, 2.95<M(ee)<3.20; 3 tracks: Δp_z>-2.5, MX within 4σ; 4 tracks: proton, -0.7<Δp_z<0.5, |MX^2|<0.036
e'e$^+$e$^-$ (proton missing)
e'e$^+$e$^-$p' (all detected)
C_{rad}
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
4.15
4.20
4.25
4.30
4.35
4.40
4.45
W (GeV), Q^2 0.02-0.12
M. Tenorio: 0.88 x 0.85
method 2 nominal cuts, inb
method 2 effective cuts, inb
method 2 flat
method 2 nominal cuts, outb
method 2 effective cuts, outb
generated 10M, generation acceptance only (normalisation corrected)
method 1, inb
method 1, outb

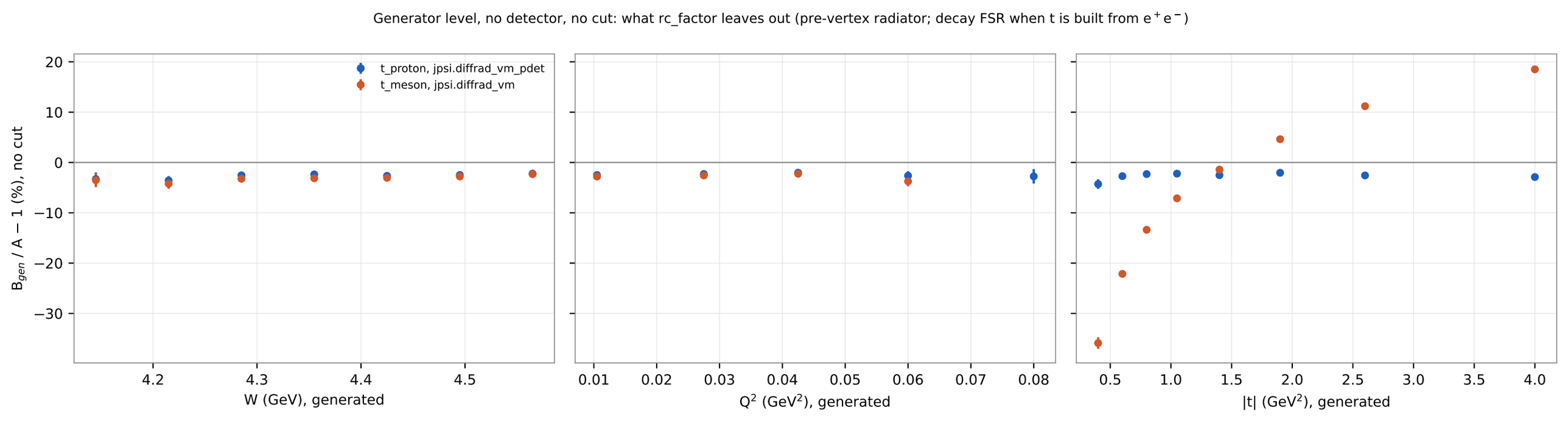

Decomposition of method 1 − method 2, e'e⁺e⁻ (proton missing)
F18 inb, nominal v 0.427
F18 outb, nominal v 0.418
% of C_rad
W (GeV)
pre-vertex radiator, hard part
decay FSR: effective edge K50
production: effective v50
migration: rec − gen bins
residual: detector response (PID, |p|)
(info) acceptance weighting A − A_flat
total: method 1 / method 2 (nominal) − 1


Decomposition of method 1 − method 2, e'e⁺e⁻p' (all detected)
F18 inb, nominal v no cut
F18 outb, nominal v no cut
% of C_{rad}
W (GeV)
pre-vertex radiator, hard part
decay FSR: effective edge K_{50}
production: effective v_{50}
migration: rec − gen bins
residual: detector response (PID, |p|)
(info) acceptance weighting A − A_{flat}
total: method 1 / method 2 (nominal) − 1

Method 2 has no torus in it: any inbending/outbending difference of method 1 is acceptance




χ^2 / cells
4×10^0
3×10^0
2×10^0
10^0
6×10^{-1}
0.0
0.2
0.4
0.6
0.8
1.0
one v for every cell (GeV^2)
3trk inb maps, 44 cells
3trk outb maps, 44 cells
4trk inb maps, 42 cells
4trk outb maps, 31 cells
cells
0
2
4
6
8
v at which method 2 equals method 1, cell by cell (GeV^2)
3trk inb: 29 of 44
3trk outb: 22 of 44
4trk inb: 37 of 42
4trk outb: 19 of 31


Generator level, no detector, no cut: what rc_factor leaves out (pre-vertex radiator; decay FSR when t is built from e⁺e⁻)
t_proton, jpsi.diffrad_vm_pdet
t_meson, jpsi.diffrad_vm
B_gen / A − 1 (%), no cut
20
10
0
−10
−20
−30
4.2
4.3
4.4
4.5
W (GeV), generated
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Q² (GeV²), generated
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
|t| (GeV²), generated